FLIGHT TELEMETRY SHEET MISSION: ASTRA-GDS DIGITAL TWIN

NODE-ALPHA (undefined)

Ground Network: **ISRO ISTRAC (8 Stations Active)** | Telemetry Engine: **Deterministic v2.4 (20 Hz HIL)**

TIMESTAMP: **2026-09-11 18:22:42 UTC**
MISSION ELAPSED: **T+00:00:06**
OVERALL HEALTH: **99% [NOMINAL OPS]**

ORBITAL EPHEMERIS

Altitude:	415.88 km
Velocity:	7.665 km/s
Coordinates:	0.0000°, 0.0000°
Orbit Rev:	#1420
Solar Illum:	UMBRA ECLIPSE (0%)

ATTITUDE & DYNAMICS (6-DOF)

Pitch (θ):	0.00°
Roll (φ):	0.00°
Yaw (ψ):	0.00°
Nadir Error:	0.18°
ADCS Mode:	EARTH_POINTING [FINE]

HARDWARE & GROUND LINK

HIL Testbed:	SIMULATED (Ready for ESP32)
Primary Pass:	ISTRAC Bengaluru
Slant Range:	7409 km
RF Quality:	-105.0 dBm (SNR 4.5 dB)
Anomaly Index:	08 / 100 (LOW)

SPACECRAFT SUBSYSTEM TELEMETRY MATRIX

FRAME VERIFICATION: CRC-32 VALID

Subsystem	Monitored Parameter	Live Measured Value	Operating Limit	Status
EPS (Electrical Power)	Solar Gen / Battery SOC	17.0 W | 95.0% (8.29V)	> 7.0V / > 20% SOC	NOMINAL
ADCS (Attitude Control)	Pointing Error / Rates	P: 0.0° | R: 0.0° | Err: 0.18°	Error < 1.0°	NOMINAL
TCS (Thermal Control)	Internal Electronics / Skin Temp	Internal: 31.5°C | Skin: -12.3°C	-20°C to +60°C	NOMINAL
OBC (Flight Computer)	CPU Load / RAM Usage / SEU Faults	CPU: 22.0% | RAM: 42.1% | SEU: 0	CPU < 85% | SEU = 0	NOMINAL
COMMS (RF Link)	Downlink RSSI / SNR Quality	-105.0 dBm | SNR: 4.5 dB (437.450 MHz)	RSSI > -120 dBm	NOMINAL
PAYLOAD (Scientific Imager)	Multispectral Sensor / Exposure State	ACTIVE | 8.4 W | Swath: 45.2 km	Max 12.0 W	ACTIVE

ACTIVE MISSION ALERTS

- [COMMS] undefined
- [EPS] undefined

RECENT OPERATIONAL TELEMETRY & TC EVENTS

[23:52:37.315] SYSTEM ISpA / IISc 2026 ASTRA-GDS Digital Twin Console initialized.
[23:52:37.319] SYSTEM Telemetry Provider started [MODE: DIGITAL TWIN / DETERMINISTIC SIMULATOR]
[23:52:37.321] SYSTEM Hardware Gateway: Standby / Extensible Adapter prepared.
[23:52:37.321] AOS LINK Ground Pass Link active with ISRO ISTRAC Bengaluru Ground Station.